Building_H1_reference

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This R Markdown document describes the construction of a correctly formatted library reference file from the hCRISPRi v2 H1 library for use with MAGeCK count. The starting hCRISPRi v2 library csv file was downloaded from addgene: https://www.addgene.org/pooled-library/weissman-human-crispri-v2/ (https://www.addgene.org/pooled-library/weissman-human-crispri-v2/)

Load raw reference library

```
# Read file using first row as header
ogref <- read.csv(
  "data_raw/hCRISPRiv2_libr.csv",
  stringsAsFactors = FALSE,
  check.names = TRUE
)

# Look at column names
colnames(ogref)
```

```
##  [1] "sgID"                          "gene"
##  [3] "transcript"                    "protospacer.sequence"
##  [5] "selection.rank"                "predicted.score"
##  [7] "empirical.score"               "off.target.stringency"
##  [9] "CRISPRi.v2.1.predicted.score" "Sublibrary"
## [11] "Sublibrary.half"
```

```
# Remove explanatory rows and empty rows
ogref <- ogref %>%
  filter(!is.na(sgID), sgID != "") %>%
  filter(!grepl("^\\[", sgID)) %>%
  filter(sgID != "sgID")

dim(ogref)
```

```
## [1] 209070     11
```

```
head(ogref)
```

```
##                      sgID gene transcript protospacer.sequence selection.rank
## 1 A1BG_-_58858617.23-P1 A1BG         P1 GGAGACCCAGCGCTAACCAG              1
## 2 A1BG_-_58858788.23-P1 A1BG         P1 GGGGCACCCAGGAGCGGTAG              2
## 3 A1BG_+_58858964.23-P1 A1BG         P1 GCTCCGGGCGACGTGGAGTG              3
## 4 A1BG_-_58858630.23-P1 A1BG         P1 GAACCAGGGGTGCCCAAGGG              4
## 5 A1BG_+_58858549.23-P1 A1BG         P1 GGCGAGGAACCGCCCAGCAA              5
## 6 A1BG_-_58858950.23-P1 A1BG         P1 GGCAGCGCAGGACGGCATCT              6
##   predicted.score empirical.score off.target.stringency
## 1     1.008815777                                     0
## 2     0.901176456                                     0
## 3     0.836188025                                     0
## 4     0.827551129                                     0
## 5     0.775395081                                     0
## 6     0.730457326                                     0
##   CRISPRi.v2.1.predicted.score Sublibrary Sublibrary.half
## 1                  0.685071255         h3            Top5
## 2                  0.782792993         h3            Top5
## 3                  0.870836555         h3            Top5
## 4                  0.590667525         h3            Top5
## 5                  0.492280233         h3            Top5
## 6                  0.706643048         h3           Supp5
```

Clean up file and filter for H1 Sublibrary. The H1 library was filtered to retain Top5 guides per gene.

```
unique(ogref$Sublibrary)
```

```
## [1] "h3" "h2" "h7" "h6" "h4" "h1" "h5"
```

```
h1 <- ogref %>%
  filter(grepl("^h1", Sublibrary, ignore.case = TRUE))
n_guides <- nrow(h1)
n_genes  <- length(unique(h1$gene))
n_guides
```

```
## [1] 26050
```

```
n_genes
```

```
## [1] 2319
```

```
# identify negative controls
neg <- h1 %>% filter(grepl("negative_control", gene, ignore.case = TRUE))
# keep top5 guides only
h1_top5 <- h1 %>%
  filter(Sublibrary.half == "Top5")
nrow(h1_top5)
```

```
## [1] 13025
```

```
length(unique(h1_top5$gene))
```

```
## [1] 2319
```

```
# check how many negative controls
neg_top5 <- h1_top5 %>% filter(grepl("negative_control", gene, ignore.case = TRUE))
nrow(neg_top5)
```

```
## [1] 250
```

Identify duplicate protospacer sequences

```
unique_seqs <- length(unique(h1_top5$protospacer.sequence))
dup_seqs <- h1_top5 %>%
  group_by(protospacer.sequence) %>%
  filter(n() > 1) %>%
  arrange(protospacer.sequence)
unique_seqs
```

```
## [1] 13008
```

```
length(unique(dup_seqs$protospacer.sequence))
```

```
## [1] 17
```

build a reference file MAGeCK can recognise MAGeCK expects 3 columns: sgRNA_ID Sequence Gene

```
mageck_lib <- h1_top5 %>%
  select(
    sgRNA_ID = sgID,
    Sequence = protospacer.sequence,
    Gene = gene
  ) %>%
  mutate(Sequence = toupper(Sequence))
# check no lowercase letters remain
any(grepl("[a-z]", mageck_lib$Sequence))
```

```
## [1] FALSE
```

```
#check number of guides
nrow(mageck_lib)
```

```
## [1] 13025
```

```
#check number of genes
length(unique(mageck_lib$Gene))
```

```
## [1] 2319
```

```
#export and save
write.table(
  mageck_lib,
  file = "results/fixed_newformattedH1ref.txt",
  sep = "\t",
  quote = FALSE,
  row.names = FALSE)
```